Linear regression and random forest regressor analysis of socioeconomic, health, and environmental determinants of life expectancy

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Introduction

This is an analysis based on data sourced from the World Health Organization's Life Expectancy dataset, publicly available on Kaggle. The study examines the factors associated with life expectancy across countries, utilizing data spanning several years to uncover critical determinants of health outcomes and socioeconomic development.

The original dataset contains variables related to health indicators (e.g., BMI, HIV/AIDS prevalence), economic metrics (e.g., GDP, percentage expenditure on health), and social determinants (e.g., schooling, status as developed or developing). This analysis focuses on assessing the relationships between these variables and life expectancy, providing a deeper understanding of global health disparities.

The analysis has three primary goals:

- Explore the statistical properties of the dataset: Are the variables normally distributed, and are there significant outliers that require adjustments?
- Evaluate the predictive power of different models. How do the results of linear regression compare to those of random forest in identifying key determinants of life expectancy?
- Identify key drivers of life expectancy: Which factors (e.g., health, education, or economic indicators) most significantly influence life expectancy?

Through this analysis, the study aims to provide actionable insights into the global health factors that influence life expectancy and offer guidance for future policy development.

Background

Life expectancy is a critical measure of a population's overall health, reflecting the combined influence of social, economic, and healthcare factors. The global disparities in life expectancy have been extensively documented, with significant gaps between high-income and low-income countries driven by varying access to healthcare, nutrition, and education (World Health Organization [WHO], 2020).

Research has highlighted the role of socioeconomic factors, such as income inequality and education, in shaping health outcomes. Marmot et al. (2012) demonstrated that higher levels of education and income are strongly correlated with improved life expectancy, particularly in countries with robust public health systems. Additionally, factors like government expenditure on health, measured as a percentage of gross domestic product (GDP), have been identified as significant predictors of better health outcomes in both developed and developing nations (Fan & Savedoff, 2014).

Health-related behaviors and disease prevalence also play critical roles. The global burden of diseases like HIV/AIDS, malnutrition, and non-communicable diseases (NCDs) such as heart disease and diabetes has contributed to stark disparities in life expectancy. For instance, Deaton (2013) found that countries with higher rates of NCDs experienced stagnating or declining life expectancy, while targeted public health interventions in low-income regions successfully addressed communicable diseases and childhood malnutrition, yielding significant improvements in survival rates.

Environmental factors and urbanization further complicate the picture. Access to clean water, improved sanitation, and reductions in air pollution have been linked to increases in life expectancy, particularly in urban areas (Prüss-Ustün et al., 2019). However, the rapid urbanization of low-income countries has also introduced challenges, such as overcrowding and inadequate infrastructure, which can exacerbate health inequities (United Nations, 2020).

Methods

This study analyzes a dataset from the World Health Organization (WHO) that includes 2,938 observations, representing data from multiple countries over the years 2000 to 2015. The analysis employs multiple linear regression and random forest models to measure and compare the associations between life expectancy and various socioeconomic, health, and environmental factors.

Data

The dataset, originally compiled by the WHO, is publicly available on Kaggle under the title "Life Expectancy (WHO)" and includes variables collected from official health and population reports. Each observation represents data for one country in a specific year, with variables spanning a range of indicators related to health, economic conditions, and demographics. The dataset covers a total of 193 countries, providing a comprehensive view of global health trends during the study period. After cleaning and preprocessing, the final dataset used for analysis contains no missing values, as missing entries were handled through a combination of interpolation and imputation techniques.

Analysis

This analysis utilizes ordinary least squares (OLS) regression and random forest modeling to examine the relationships between life expectancy and multiple independent variables.

Variables

Of the 22 variables included in the dataset, the following are considered in this analysis:

Table 1: Variables

Life expectancy	The target variable, representing the average age at death in years for a given population.
Status	Whether a country is classified as "Developing" or "Developed" by the WHO; encoded as 0 = Developing, 1 = Developed.
Adult Mortality	Probability of dying between 15 and 60 years per 1,000 population.
infant deaths	Number of infant deaths per 1,000 live births.
Alcohol	Per capita alcohol consumption (15+ years) measured in liters of pure alcohol.
percentage expenditure	Expenditure on health as a percentage of GDP per capita.
Hepatitis B	Immunization coverage of one-year-olds against Hepatitis B, expressed as a percentage.
Total expenditure	General government expenditure on health as a percentage of total government expenditure.
Measles	Number of reported cases per 1,000 population.
BMI	Average Body Mass Index of the population.
Polio	Immunization coverage of one-year-olds against Polio, expressed as a percentage.
Diphtheria	Immunization coverage of one-year-olds against Diphtheria, Tetanus, and Pertussis (DTP3), expressed as a percentage.
HIV/AIDS	: Number of deaths due to HIV/AIDS per 1,000 live births in children aged 0–4 years.
Population	: Total population of the country.
thinness 1-19 years	Prevalence of thinness among children and adolescents aged 10–19 years, expressed as a percentage.
Schooling	Average number of years of schooling for the population.

Results

Descriptives

Table 1 provides a summary of the descriptive statistics for the dataset, which includes observations spanning multiple countries and years from 2000 to 2015. All missing values were addressed through a combination of methods, including linear interpolation within each country grouping followed by mean imputation. The dataset is complete, with no missing values, ensuring consistency across all variables.

Table 2: Descriptive Statistics

Variable	Min	Median	Mean	Max	S.D.
Year	2000	2007.5	2007.5	2015	4.61056
Life expectancy	36.3	72.1	69.224932	89	9.523867
Adult Mortality	1	144	164.796448	723	124.2920
infant deaths	0	3	30.407445	1800	118.1144
Alcohol	0.01	3.65	4.528796	17.87	4.05853
percentage	0	65.611455	740.321185	19479.9116	1990.930
Hepatitis B	1	88	75.815374	99	27.88081
Measles	0	17	2427.85587	212183	11485.97
BMI	1	43	38.235394	77.6	19.85018
Polio	3	93	82.304986	99	23.62616
Total expenditure	0.37	5.78	5.931609	17.6	2.489875
Diphtheria	2	93	82.321416	99	23.62957
HIV/AIDS	0.1	0.1	1.747712	50.6	5.085542
Population	34	3589254	12743010	1293859000	5390820
thinness 1-19 years	0.1	3.4	4.850622	27.7	4.396597
Schooling	0	12.1	11.999639	20.7	3.253691

The mean life expectancy in the dataset is approximately 69.22 years, with a standard deviation of 9.52 years, indicating moderate variation across countries. The minimum recorded life expectancy is 36.3 years, while the maximum is 89 years. Adult mortality rates show a mean of

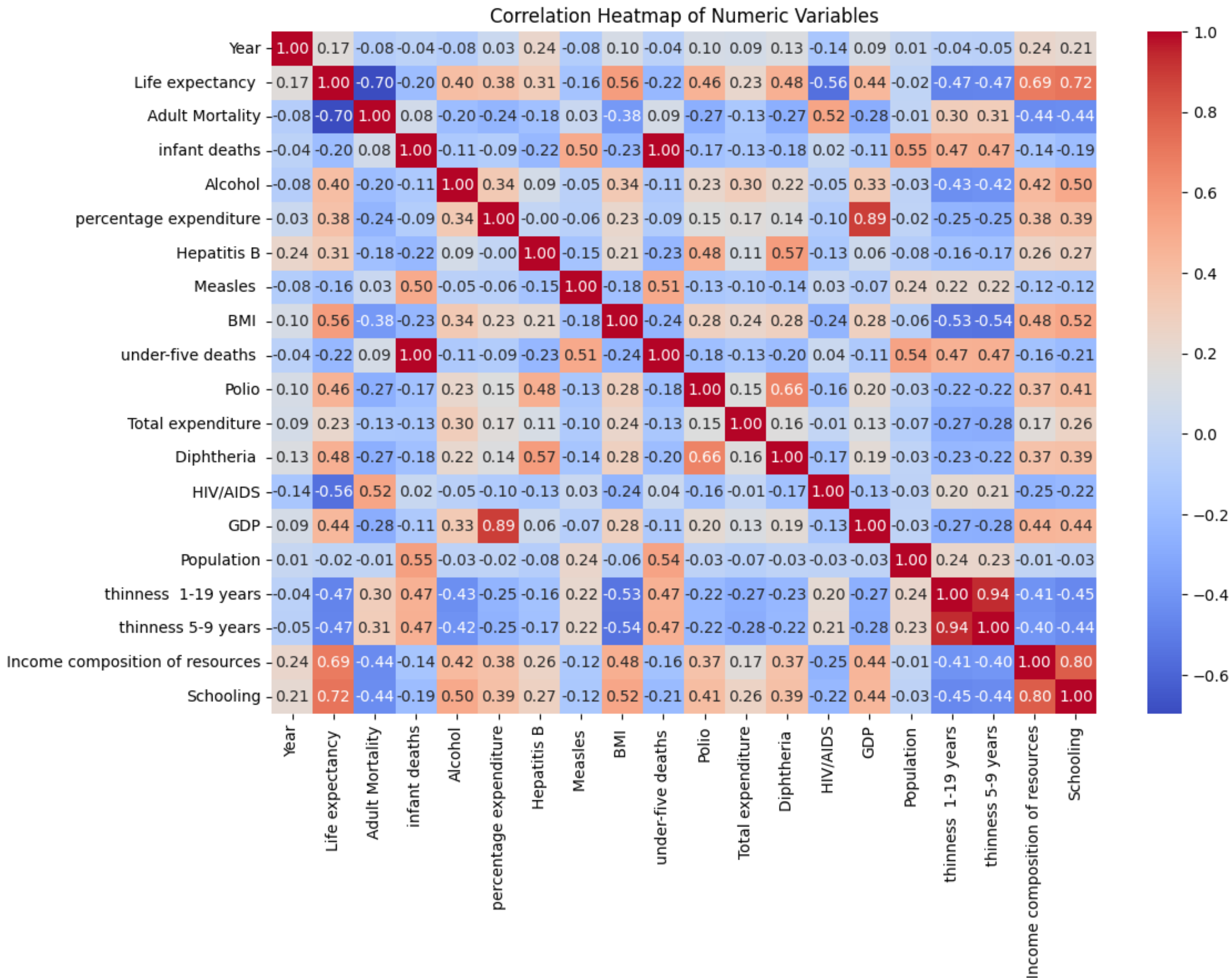
164.8 deaths per 1,000 adults, with some countries experiencing as few as 1 death per 1,000 and others as many as 723.

On average, countries report 30.4 infant deaths per 1,000 live births, though the standard deviation of 118.1 highlights significant disparities. Alcohol consumption per capita averages 4.53 liters, with values ranging from 0.01 to 17.87 liters. Economic factors also show considerable variation. Government expenditure on health, expressed as a percentage of GDP, averages 740.32%, but this variable has a large standard deviation of 1990.93, reflecting outliers among countries with extreme values. Similarly, GDP ranges widely, with a mean of $7,378 per capita but a maximum of $119,172. Health interventions, such as immunization coverage, indicate relatively high averages. For example, Hepatitis B immunization coverage is 75.82%, and Polio coverage is 82.30%. The average Body Mass Index (BMI) for the population is 38.24, with a minimum of 1.0 and a maximum of 77.6. On average, the prevalence of thinness among adolescents is 4.85%, while the average years of schooling across all countries is approximately 12 years.

Correlations

The correlation heatmap visualizes the relationships between numeric variables in the dataset, highlighting key associations and potential multicollinearity issues. Life expectancy shows significant positive correlations with variables such as schooling (0.72), income composition of resources (0.69), and health-related factors like immunization coverage (e.g., Polio: 0.56). On the other hand, it exhibits negative correlations with variables such as adult mortality (0.70) and HIV/AIDS (0.56). These patterns reflect the complex interactions between socioeconomic, health, and environmental factors in determining life expectancy.

Figure 1: Correlation Heatmap



A closer examination of the correlations reveals potential multicollinearity concerns. For example, income composition of resources has a very high correlation with schooling (0.80), indicating redundancy between these variables. Including both in the regression model could lead to inflated standard errors and unreliable coefficient estimates. Similarly, under-five deaths and infant deaths are nearly identical in their explanatory power, with a correlation of 0.99. This makes it unnecessary to include both variables in the model. Another instance of redundancy is observed between thinness 1-19 years and thinness 5-9 years, which are strongly correlated (0.94), suggesting that they capture the same underlying pattern.

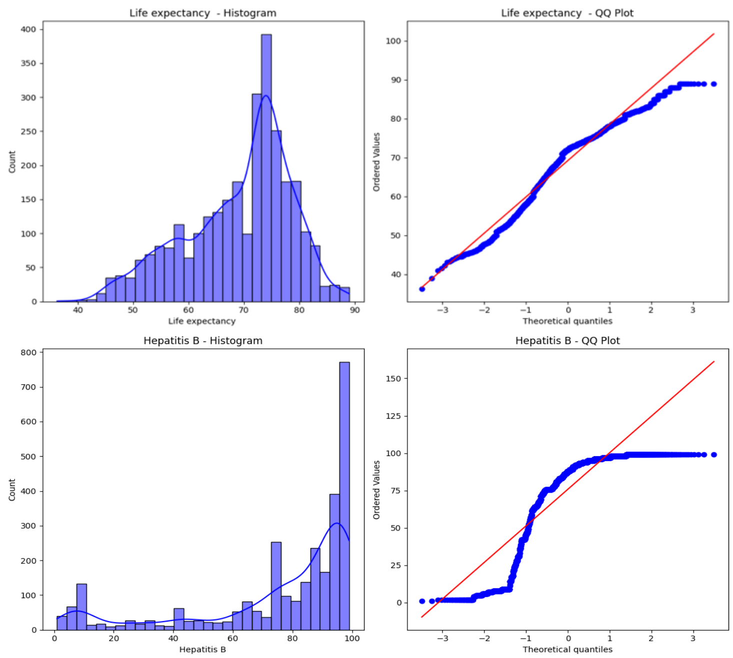
Given these findings, several variables were removed from the final model to address redundancy and improve efficiency. Income composition of resources, despite its strong correlation with life expectancy, was excluded due to its high collinearity with schooling. Schooling was retained as it serves as a more direct indicator of human capital and development. Under-five deaths was removed in favor of infant deaths, which provides similar explanatory contributions while avoiding multicollinearity issues. Finally, thinness 5-9 years was excluded because thinness 1-19 years captures a broader age range and offers a more comprehensive perspective. These adjustments ensure that the final model retains only the most informative and unique predictors, improving its reliability and interpretability.

Distributional Analysis

The dataset's variables were assessed using histograms, QQ plots, skewness, kurtosis, and the Shapiro-Wilk test to evaluate their distributional properties and identify deviations from normality.

Several variables demonstrate approximate symmetry or mild skewness, such as Life expectancy (Skewness = − 0.638) and Schooling (Skewness = − 0.600), indicating distributions that are close to normal. Other variables show substantial deviations from normality, with extreme skewness and kurtosis values. For instance, Population (Skewness = 18.00, Kurtosis = 381.69) and infant deaths (Skewness = 9.77, Kurtosis = 115.46) exhibit highly skewed and heavy-tailed distributions. Similarly, variables like Measles and HIV/AIDS display significant outliers and positive skewness (Skewness = 9.42 and 5.38).

Figure 2: Histograms and QQ plots



Interestingly, some health-related indicators, such as Polio (Skewness = − 2.07) and Diphtheria (Skewness = −2.08), exhibit negative skewness, indicating that most countries report high immunization rates with only a few outliers on the lower end. Economic variables like GDP (Skewness = 3.48) and percentage expenditure (Skewness = 4.64) are positively skewed, reflecting the unequal distribution of wealth and health spending globally. These variables also show high kurtosis values, indicating the presence of extreme outliers.

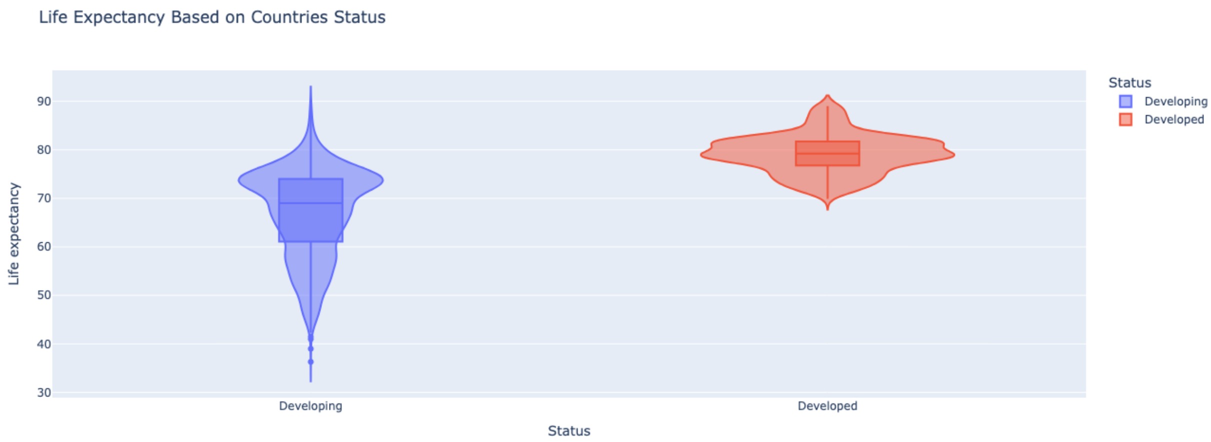
The results reveal substantial diversity in the distributions of the variables, reflecting the inherent differences across countries in terms of health, socioeconomic conditions, and population characteristics. This diversity is an essential feature of the dataset, as it captures the variations among countries rather than assuming uniformity. As a result, the data do not require further transformations or adjustments for normality, as the goal is to analyze these variations rather than standardize them.

To provide an overview of the distributional properties, a selection of histograms and QQ plots is included in the main report to highlight key trends and observations. The complete set of histograms and QQ plots is provided in the appendix for reference.

Visualizations

The violin plot above illustrates the distribution of life expectancy for countries categorized as "Developing" and "Developed."

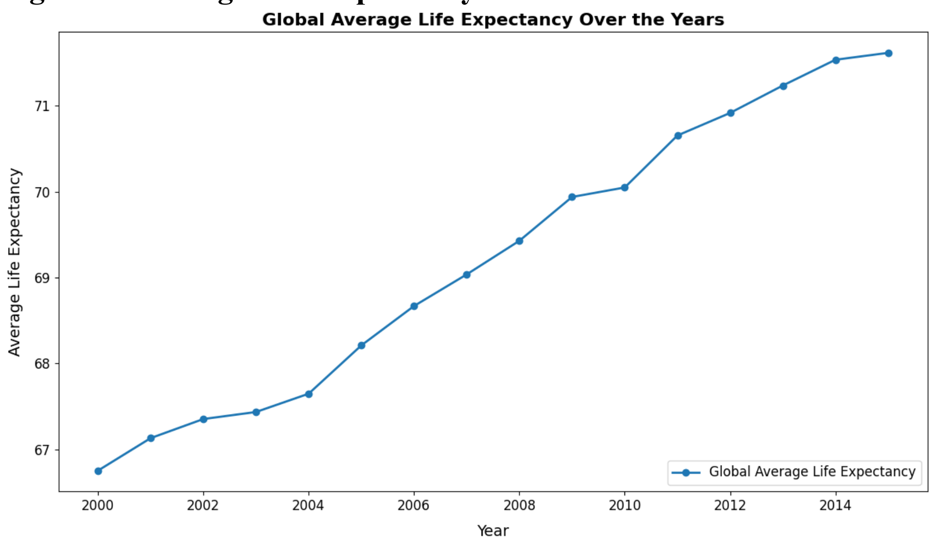
Figure 3: Violin Plot



For developed countries, the distribution of life expectancy is concentrated at higher values, with the median exceeding 75 years. The interquartile range is relatively narrow, indicating less variation in life expectancy within this group. The boxplot embedded within the violin plot confirms this observation, as it shows a compact range of values with minimal outliers. This suggests that developed countries consistently achieve high life expectancy due to better access to healthcare, nutrition, and other social determinants of health.

In contrast, developing countries exhibit a broader and more variable distribution of life expectancy. The median life expectancy for this group is significantly lower, approximately 65 years, and the IQR is wider. Additionally, the tails of the distribution extend further, indicating greater disparity in health outcomes among developing countries. This variability may be attributed to uneven access to healthcare, economic challenges, and differences in infrastructure and policy implementation across countries.

Figure 4: Average Life Expectancy Plot

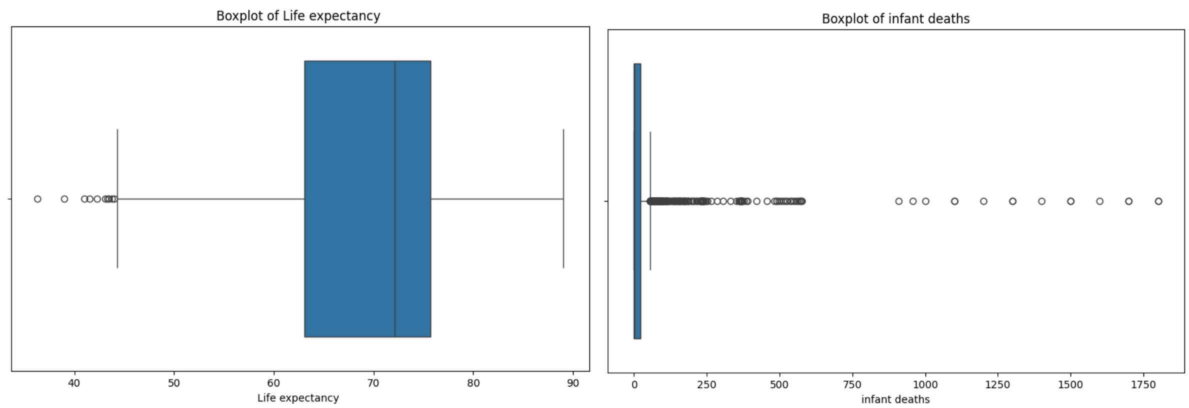


The line plot illustrates the global average life expectancy trends from 2000 to 2015. The chart reveals a steady and consistent increase in life expectancy during this period, rising from approximately 67 years in 2000 to over 71 years by 2015.

Outliers

The boxplots for variables such as Life Expectancy and Infant Deaths illustrate the presence of numerous outliers across the dataset. These outliers highlight the inherent diversity in the dataset, as it includes data from countries with vastly different socioeconomic and health conditions. The outliers are an important reflection of real-world disparities and do not necessarily represent errors or noise in the data. Instead, they provide valuable insights into the extreme values that characterize specific regions or time periods.

Figure 5: Boxplot



Most of the outliers were retained to preserve the variability and richness of the data. However, 10 outliers in Life Expectancy were removed due to their extreme deviation from the overall trend.

Decision Criteria

The decision criteria for this analysis are based on identifying the key drivers of Life Expectancy and understanding their relative importance. The selection of variables is guided by their correlation strength and practical significance. Variables such as Schooling, Adult Mortality, Income composition of resources, and Polio immunization coverage are prioritized due to their strong correlation with life expectancy. Additionally, the diversity in the dataset is considered, ensuring that the criteria reflect global health disparities between developed and developing countries. The decision criteria aim to balance statistical rigor with practical interpretability, focusing on variables that are both statistically significant and policy relevant.

Assumptions

The analysis assumes that the dataset accurately reflects real-world health and socioeconomic conditions across countries. It is further assumed that the relationships between variables are linear and consistent across time and regions, barring specific interaction effects. For the regression models, normality of residuals, homoscedasticity, and minimal multicollinearity are assumed to hold, ensuring reliable coefficient estimates. Additionally, it is presumed that the imputation methods used for missing data do not introduce significant bias or alter the underlying relationships in the data. These assumptions provide a foundation for building robust and interpretable models to guide actionable insights.

Model 1: Linear Regression Model

The Model 1 examines the relationship between Life Expectancy and several socioeconomic and health-related predictors. The R^2 value of 0.802 indicates that approximately 80.2% of the variation in life expectancy can be explained by the independent variables included in the model

Table 3: Regression Results

Variable	Coefficient	P-Value	
const	53.8784	0.0000	***
Schooling	0.944592	0.0000	***
HIV/AIDS	-0.48946	0.0000	***
Adult Mortality	-0.02075	0.0000	***
Diphtheria	0.047896	0.0000	***
BMI	0.049089	0.0000	***
percentage expenditure	0.000306	0.0000	***
Status	1.662484	0.0000	***
Polio	0.024542	0.0000	***
Measles	-3.3E-05	0.0000	***
thinness 1-19 years	-0.07998	0.0010	**
Population	3.4E-09	0.0487	*
Total expenditure	0.066499	0.0495	*
Alcohol	0.031264	0.2250	
infant deaths	-0.0008	0.4043	
Hepatitis B	-0.00155	0.6574	

Among the predictors, Schooling emerges as the strongest positive association with life expectancy. Similarly, variables such as Diphtheria, BMI, and Polio show significant positive impacts on life expectancy, emphasizing the importance of immunization programs and general health metrics in increasing life spans.

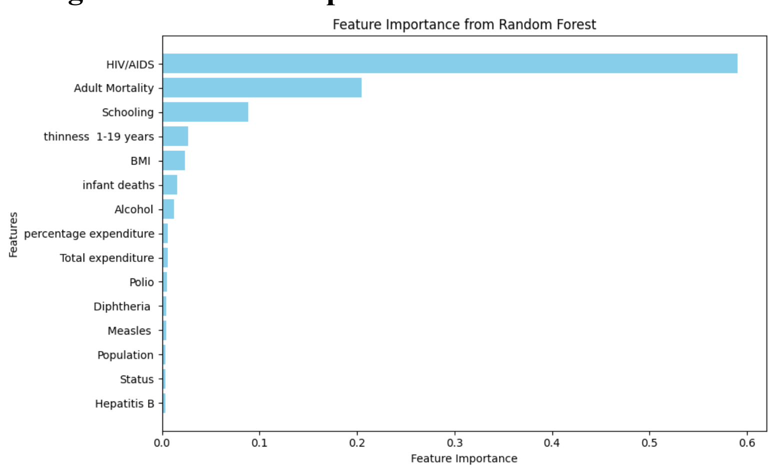
Conversely, several variables demonstrate significant negative associations with life expectancy. HIV/AIDS exerts a substantial negative effect, reflecting the critical need to address the prevalence of infectious diseases in many regions. Similarly, Adult Mortality negatively impacts life expectancy.

Moderately significant predictors include Population and Total expenditure. Non-significant variables, such as Alcohol, infant deaths, and Hepatitis B, indicate that these factors may not directly influence life expectancy or that their effects are mediated by other included predictors.

Model 2: Random Forest Regressor Model

The second model utilizes a Random Forest Regressor to predict Life Expectancy using the same independent variables as the linear regression model. The Random Forest model achieves an R^2 of 0.958, indicating that it explains approximately 95.8% of the variance in life expectancy. The RMSE (Root Mean Square Error) of 1.96 demonstrates the model's strong predictive accuracy.

Figure 6: Feature Importance



The feature importance analysis highlights which predictors play the most significant roles in determining life expectancy. HIV/AIDS emerges as the most influential variable, with an importance score of 0.590. Adult Mortality follows, with an importance score of 0.205, emphasizing its direct association with longevity. Schooling ranks third (0.089), reaffirming its substantial, albeit less pronounced, role compared to HIV/AIDS and adult mortality. Other notable predictors include thinness 1-19 years (0.027), BMI (0.024), and infant deaths (0.016), all of which contribute to understanding life expectancy at varying levels. Lower-ranked features, such as Diphtheria (0.004), Measles (0.004), and Population (0.004), exhibit minimal importance in the model.

Table 4: Random Forest Results

Feature	Importance
HIV/AIDS	0.590487
Adult Mortality	0.204857
Schooling	0.088909
thinness 1-19 years	0.027226
BMI	0.023597
infant deaths	0.015951
Alcohol	0.012125
percentage expenditure	0.006327
Total expenditure	0.005914
Polio	0.004883
Diphtheria	0.004404
Measles	0.004346
Population	0.003877
Status	0.003827
Hepatitis B	0.00327

While linear regression identifies Schooling as the most significant positive predictor, Random Forest ranks HIV/AIDS and Adult Mortality higher in importance, providing a different perspective on key drivers of life expectancy.

Model 3: Interaction Effects Between Status, Schooling, and Immunization

This model introduces interaction terms to explore the combined effects of Status (Developed vs. Developing), Schooling, and Immunization (Polio) on life expectancy. Two interaction terms are added: Status × Schooling and Status × Polio. These terms aim to capture how the influence of schooling and immunization varies between developed and developing countries.

Table 5: Regression Results

Variable	Coefficient	P-Value	
const	53.35217	0.0000	***
Schooling	1.011695	0.0000	***
HIV/AIDS	-0.48834	0.0000	***
Adult Mortality	-0.02042	0.0000	***
Diphtheria	0.04708	0.0000	***
BMI	0.04701	0.0000	***
percentage expenditure	0.000361	0.0000	***
Status	14.44089	0.0000	***
Status_Schooling	-0.58107	0.0000	***
Polio	0.025615	0.0000	***
Measles	-3.3E-05	0.0000	***
thinness 1-19 years	-0.07901	0.0000	**
Status_Immunization	-0.04228	0.0013	*
Population	3.25E-09	0.0162	
Total expenditure	0.0607	0.0581	
Hepatitis B	-0.00366	0.0718	
infant deaths	-0.00075	0.2945	
Alcohol	0.009871	0.4327	

Notably, the variable Status itself has a large and statistically significant positive coefficient (14.44), indicating that developed countries, on average, have significantly higher life expectancy. However, the significant interaction terms underline the complexity of the relationships, highlighting how development status interacts with other critical factors like education and healthcare.

The regression results for this model reveals several important findings. The model achieves a robust fit, with significant coefficients for the interaction terms and a notable impact on life expectancy. The interaction term Status_Schooling has a coefficient of - 0.581 and a p-value of 1.001e-08, indicating a strong and statistically significant negative relationship. This suggests that schooling appears to have a stronger effect in developing countries compared to developed ones. Similarly, the interaction term Status_Immunization shows a coefficient of - 0.042 with a p-value of 0.016, indicating statistical significance. This result suggests that the effectiveness of immunization on life expectancy is also moderated by development status. The negative coefficient implies that immunization has a more pronounced positive effect on life expectancy in developing countries than in developed ones.

In summary, this model demonstrates that the effects of schooling and immunization on life expectancy are significantly influenced by a country's development status. These findings emphasize the importance of tailored interventions that consider a country's development context, particularly in areas such as education and healthcare.

Conclusion

This report explored the factors influencing life expectancy across countries using data from the World Health Organization's Life Expectancy dataset. Employing multiple analytical approaches, including linear regression, random forest modeling, and interaction-effect analysis, the findings highlight the critical interplay of socioeconomic, health, and environmental variables in shaping life expectancy outcomes globally.

The analysis confirms the significant role of key predictors such as schooling, immunization coverage, and healthcare expenditure in extending life expectancy. Notably, schooling consistently emerged as a strong positive factor, underscoring the importance of education in fostering healthier populations. On the other hand, adverse factors such as HIV/AIDS prevalence and adult mortality were identified as substantial negative drivers, emphasizing the need for targeted interventions to address these challenges.

Comparative modeling provided valuable insights, with random forest analysis reinforcing the importance of health-related predictors such as HIV/AIDS and adult mortality, while interaction-effect models revealed nuanced relationships influenced by development status. These interactions demonstrate that factors like schooling and immunization have stronger impacts in developing countries compared to developed ones, reflecting the unequal distribution of resources and healthcare systems.

This report also underscores the complexity of global health disparities. While developed countries consistently achieve higher life expectancy due to stable infrastructures and robust healthcare systems, the variability among developing countries highlights the pressing need for tailored strategies that address localized challenges. Moreover, the analysis identified key opportunities for policy action, such as increasing educational access and enhancing healthcare interventions, particularly in regions with high disease burdens and limited resources.

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Appendix:

Figure 7: Histograms and QQ plots

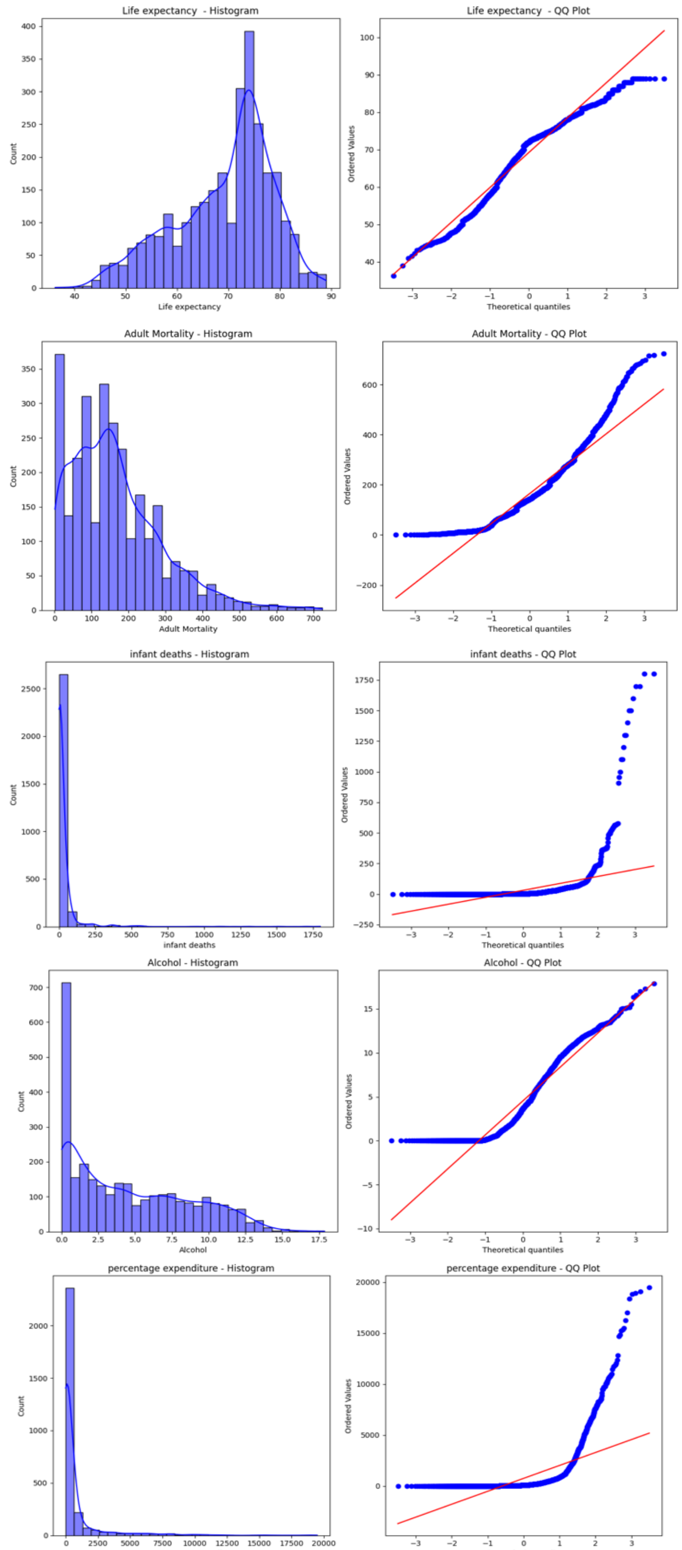

Figure 8: Histograms and QQ plots

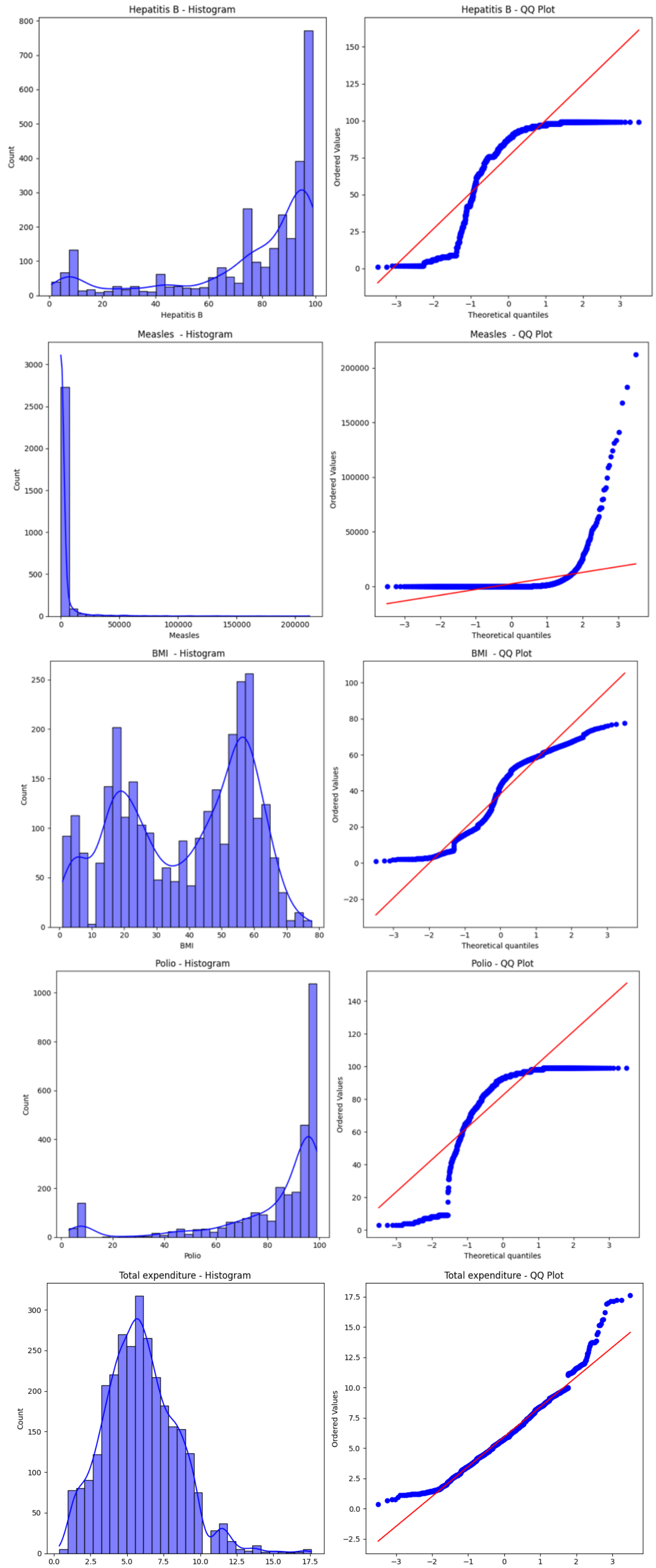

Figure 9: Histograms and QQ plots